ML FOUNDATIONS

PROJECT URL:https://github.com/faisalxy-lab/FinalProject

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INTRODUCTION

In this project, I used a dataset for the Titanic passengers, which is available in a file called trainproject.xlsx. This dataset contains demographic, behavioral, and financial data for the passengers, including: gender, age, class (Pclass), ticket price, family size, departure port, and whether they lived or not.

What questions was I trying to answer?

01 WHAT DO THE BASIC DISTRIBUTIONS OF NUMERICAL VARIABLES LOOK LIKE, AND DO THEY SHOW SIGNS OF SKEWNESS OR OUTLIERS?

02 ARE THERE ANY MISSING OR INVALID VALUES THAT IMPACT THE QUALITY OF THE DATA? AND HOW WERE THEY HANDLED?

03 WHICH VARIABLES HAVE THE MOST IMPACT ON THE PROBABILITY OF A PASSENGER SURVIVING?

CLEANING SUMMARY

In the data analysis process, several issues concerning the quality of the Titanic data were identified and addressed systematically. The following are the major issues concerning the data, and how each of them was addressed

	Missing Values	Invalid Values	Outliers	Skewness	Duplicates
PROBLEM	The column "Age" has a large number of missing values. The "Embarked" column has missing values. The "Cabin" column has a very high percentage of missing values, i.e., above 70%.	The Survived column may have illogical values other than 0 and 1. Text columns were not stored in the correct data type.	**The presence of values outside of the normal range for:** **Age** **Fare** **SibSp** **Parch**	Fare column also had high skewness (skewness = 4.78). This affected the analysis and models.	Duplicate rows.
FIX	Missing values were replaced with the median for the Age variable because of skewness, while the Embarked variable was replaced with the mode because it is categorical data. The Cabin variable was removed because of too many missing values and because it does not provide significant information for analysis.	The Survived column has been checked and verified to have no negative values. Changing the column's data type: Name, Ticket, Cabin: string Sex, Embarked, Pclass: category	I Data was visually checked using a boxplot method. Outliers were included in the analysis as they might carry important information (very costly tickets).	Using log1p() transformation for reducing skewness and improving the distribution. **After applying the log1p() transformation, the skewness is very low (0.39).**	Check if there are any duplicate rows by using df.duplicated().sum(). This will ensure that there are no duplicates that might affect the data.

FEATURE ENGINEERING SUMMARY

What new columns did I create, and why did I create them?

FamilySize

In order to calculate the size of the family accompanying the passenger (parents + siblings + himself), as this will influence survival.

fare_per_age

In order to calculate the ratio between the cost of the ticket and the age, which will provide us with an economic indicator.

fare_per_age

In order to calculate the size of the family relative to the cost of the ticket.

Fare_Family_interaction

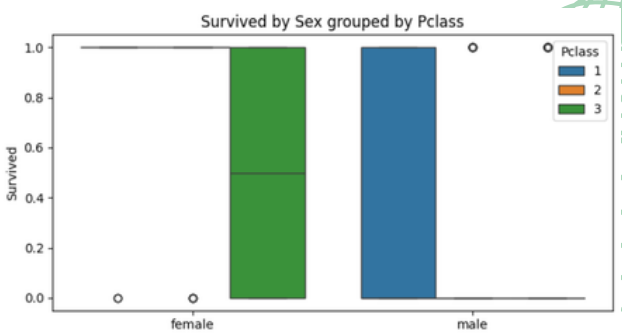
In order to link the cost of the ticket to the size of the family in one column.

Age_group

In order to divide the passengers into specific categories according to their age (child, teenager, adult, senior).

THE BEST CHART I HAVE

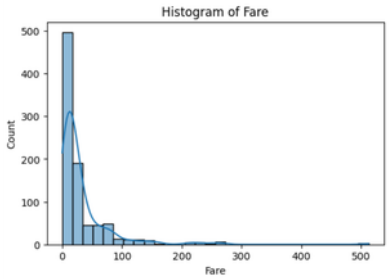
The chart also clearly highlights the strongest pattern that I was able to derive from the Titanic dataset, which showed that female passengers and those in first class had significantly higher survival rates. It was the most powerful visualization of the factors that impact survival, and that is why I have chosen it as the most meaningful visualization that I created.



WHAT ARE THE 3 MOST INTERESTING THINGS I DISCOVERED?

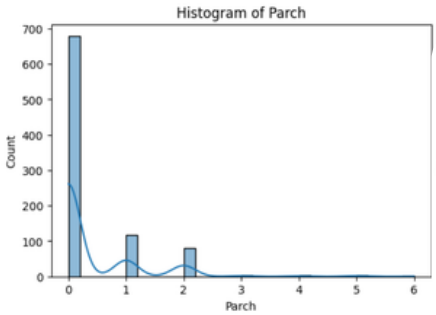
THE TICKET PRICE (FARE) IS HIGHLY DEVIANT BASED ON MANY EXTREME VALUES.

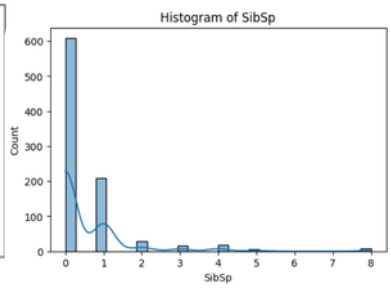
From the graph, it is clear that most passengers paid very low fares, but a few paid very high fares, going up to about 500. This skewed the distribution of the data.
The log transformation was applied for a better graph.



MOST WERE TRAVELING WITHOUT FAMILY (SIBSP AND PARCH ≈ 0)

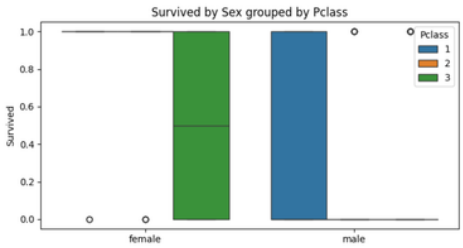
From the two graphs above, we can deduce that the majority of the passengers were traveling alone since the values for both SibSp and Parch were close to 0.
This affected the survival rate; passengers traveling alone had a lower survival rate.





THE TICKET PRICE (FARE) IS HIGHLY DEVIANT BASED ON MANY EXTREME VALUES.

The combined chart clearly shows that:
Survival rate is high for females.
Survival rate is high for First class.
Survival rate is high for both females and First class combined.



WHAT WOULD I DO NEXT IF I HAD MORE TIME?

If I had more time, I would continue my analysis by creating a complete end-to-end pipeline using the course structure as a basis. I would also explore more complex data cleaning techniques, incorporate more interaction and domain-driven features, and explore different data transformation techniques, including box Cox and robust scaling. I would also explore different families of models, compare their performance, and assess the importance of the features, among other things.